The generic structure is the free Steiner triple system on countably many generators

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Editorial note. The counted statement is the author's environment at source0.tex lines 760-761 and its proof environment opens at line 762; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

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Editorial scope. One original result of the pinned author manuscript, counted once: the generic is the free system on countably many generators (author label Feb_23_2024_4).

Allowed standard prerequisites (editorial).

- The generic structure. The generic is the model obtained by the amalgamation construction from the finite unconfined systems, with a distinguished set of vertices.. It is the structure shown to be free.
- Freely generated system. A system is freely generated by a set when that set is independent and the system is generated by it.. It is the conclusion of the statement.

- Extension of finite embeddings. Finiteness of the generating data makes every partial embedding from a finite subset extend to an embedding of the generated subsystem.. It is what forces the universal mapping property.

Proposition 0.1 Assume M is a Steiner triple system and $A \subseteq M$. The following are equivalent:

1. A is a free base of M (as a Steiner quasigroup)
2. There is a well-founded F-ordering of M over A
3. There is a well-founded HF-ordering of M and A is the set of points that are not a product of smaller elements.

Proposition 0.2 The generic is isomorphic to the free Steiner triple system with ω generators.

Proof. First note that the generic F is an STS, that is, the product of two distinct elements a_1, a_2 is defined: let b be a new vertex and let $A = \{a_1, a_2, b\}$ be an STS (so $\{a_1, a_2, b\}$ is a block). Then $A \in \mathcal{C}_{\text{un}}$. Let $B \leq F$ be such that $\{a_1, a_2\} \subseteq B$. If $a_1 \cdot a_2 \notin B$, then $a_1 \cdot a_2$ can be put on top in an HF-ordering of B and, therefore, $B \leq B(a_1 \cdot a_2)$. So there is an embedding $f : B(a_1 \cdot a_2) \to F$ over $B \supseteq \{a_1, a_2\}$. Moreover, F must be a free Steiner triple system, since it has a well-founded HF-ordering. On the other hand, the generic can be constructed by including at each stage an element which does not form a block with any previous elements. Then the corresponding HF-ordering has infinitely many elements which do not form a block with any preceding elements. By Proposition 0.1, these elements belong to a free base of F.

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